

London Underground “What-If” Simulation

Legal Considerations Document

1. Introduction

This document outlines the legal considerations associated with the development of the London Underground “What-If” Simulation. The project integrates third-party transport data obtained from Transport for London (TfL) and presents a visual representation of the London Underground network. As a result, several legal areas must be considered, including copyright, intellectual property, software licensing, data licensing and data protection.

The purpose of this document is to demonstrate how the project has been designed to minimise legal risk while ensuring lawful and responsible use of publicly available transport data.

2. Use of TfL Open Data

2.1. TfL Open Data Licence

The application uses data provided through the TfL Unified API under the TfL Open Data Licence. This licence permits the use of TfL transport data for educational, research and non-commercial purposes, subject to the conditions specified by TfL.

The data utilised within the project includes:

- Underground line information
- Station metadata
- Route connectivity information
- Live service status data
- Train arrival prediction data

This information is used to construct the network graph, support route calculation and provide real-time operational information within the simulator.

Compliance Measures:

To ensure compliance with the TfL Open Data Licence:

- TfL is acknowledged as the source of transport data within project documentation.
- All data is retrieved through official TfL API endpoints.
- Raw TfL datasets are not redistributed, sold or republished.
- Data is used solely for educational and demonstration purposes.
- Transformations performed within the application do not modify the original TfL data source.

By consuming data directly from official endpoints and avoiding redistribution of raw datasets, the project remains consistent with the intended use of TfL Open Data.

3. Copyright Considerations

3.1. London Underground Map Design

The official London Underground map is protected by copyright and design rights. Reproducing, tracing or closely imitating the schematic layout without permission may constitute copyright infringement, even where the design has been recreated digitally.

To avoid infringement, the project does not attempt to replicate the official Underground map design.

Compliance Measures:

- The application does not reproduce the official schematic Underground map.
- The network visualisation is generated programmatically from transport data.
- Station locations are derived from geographic coordinates rather than schematic placement.
- Route rendering is generated dynamically from graph relationships.
- No copyrighted TfL map images, diagrams or design assets are embedded within the application.
- The application does not attempt to replicate the distinctive layout, typography or iconography associated with the official Underground map.

This approach ensures that the visualisation constitutes an original representation of publicly available transport data rather than a derivative reproduction of a copyrighted work.

3.2. Use of Line Names and Colours

The application uses official Underground line names and line colours to support route identification and improve usability.

Compliance Measures:

- Line colours are used solely for informational and identification purposes.
- The use of colours does not recreate the overall visual appearance of the official Underground map.
- TfL logos, branding assets and proprietary design elements are not used.
- No attempt is made to imply endorsement by or affiliation with TfL.

The use of line names and colours is therefore limited to functional representation of transport information rather than brand reproduction.

4. Intellectual Property

The source code, routing algorithms, disruption simulation logic, visualisation technique and original documentation created during the project remain the intellectual property of the project authors.

Ownership of third-party resources remains with their respective owners.

This includes:

- TfL transport data
- Open-source software libraries
- External APIs and datasets

The project does not claim ownership of any third-party intellectual property incorporated within the system.

4.1. Third-Party Software and Open-Source Libraries

The application incorporates several open-source software libraries and frameworks that are distributed under permissive software licences.

Examples include:

- React
- Next.js
- Leaflet
- React-Leaflet
- Neo4j JavaScript Driver

These libraries remain the intellectual property of their respective owners and are used in accordance with their published licence terms.

The project does not modify, redistribute or claim ownership of these components beyond their intended use within the application.

5. Data Protection and Privacy

5.1. Personal Data

The application has been designed so that it does not intentionally collect, store or process personal data.

Specifically:

- No user accounts are created.
- No personally identifiable information (PII) is collected.
- No journey histories are stored.
- No behavioural tracking is performed.
- No user-generated content is transmitted to external services.

As no personal information is processed by the application, the project's obligations under the UK GDPR and the Data Protection Act 2018 are minimal.

This design decision simplifies compliance requirements while reducing privacy risks for users.

6. Liability and Disclaimer

Although the simulator incorporates live TfL data, it is not intended to function as an official journey planning service.

The application is provided for educational, analytical and demonstration purposes only.

Compliance Measures:

- A disclaimer identifies the application as an independent university project.
- The application does not claim endorsement by or affiliation with TfL.
- Route recommendations and disruption simulations are provided for demonstration purposes only.
- Users are advised to consult official TfL services when making travel decisions.
- Live data may be delayed, incomplete or temporarily unavailable depending on the status of the TfL API.

These measures help reduce the risk of users relying on the simulator as an authoritative transport information source.

7. Risk Assessment and Mitigation

Legal Risk	Mitigation Strategy
Copyright infringement	No reproduction of the official Underground schematic map
Breach of TfL licence terms	Use restricted to permitted TfL Open Data endpoints with appropriate attribution
Misuse of TfL branding	No TfL logos, typography or proprietary design assets used
Open-source licence violations	Libraries used in accordance with published licence terms
GDPR non-compliance	No personal data collected, stored or processed
Misinterpretation of travel information	Clear disclaimer and educational-use positioning

8. Conclusion

The TfL “What-If” Transport Network Simulator has been designed to comply with relevant legal, licensing and intellectual property requirements.

By avoiding reproduction of copyrighted map designs, adhering to TfL Open Data licensing conditions, respecting open-source software licences and ensuring that no personal data is processed, the project minimises legal risk while making responsible use of public transport information.

The project demonstrates how publicly available transport data can be integrated into an educational software system while respecting copyright law, licensing obligations and data protection principles.